ONLINE SUPPLEMENT

Synthetic Respondents without Ambivalence? A Population-Fidelity Audit of Large Language Model Survey Simulations

1 Reproduction Map

The supplement documents the complete empirical inputs behind the manuscript. The replication archive is organized as follows:

- `scripts/00_build_authorized_benchmark.R`: reconstructs the restricted benchmark object from locally held, authorized CGSS source files.
- `scripts/01_prepare_profiles.R`: draws the weighted, stratified profile sample and removes held-out human outcomes from the model-facing file.
- `scripts/02_run_local_llm.py`: calls the local Ollama endpoint, validates JSON outputs, records seeds and prompt hashes, and resumes incomplete runs.
- `scripts/03_evaluate_audit.R`: computes distributions, variance, correlations, profile errors, prompt sensitivity, repeated-draw variance decomposition, profile-cluster intervals, and subgroup gradients.
- `scripts/04_run_independent_items.py`: presents one item per fresh model call, repeating the complete persona without retaining conversation history.
- `scripts/05_extended_validation.R`: constructs the human-sampling reference envelopes, weighted and polychoric correlation checks, and stratified joint-donor benchmark.
- `ml/scripts/01_export_data.R` and `ml/scripts/02_train_evaluate.py`: construct the survey-trained reference models, out-of-fold predictions, temporal-transfer predictions, and profile-cluster bootstrap.
- `paper_smr/reproduce.R`: regenerates the Monte Carlo study and every main-text figure from the saved analysis outputs.

The restricted local working copy retains immutable synthetic-response logs. The conservative public archive excludes profile-level logs and respondent-derived persona files; it contains code and aggregate results. Reconstructing the full audit therefore requires authorized access to the source CGSS waves and regenerated local model calls. Regenerating calls requires Ollama 0.31.1 and the `qwen3:8b` Q4_K_M model. The recorded Ollama model digest is:

```
500a1f067a9f782620b40bee6f7b0c89
e17ae61f686b92c24933e4ca4b2b8b41
```

2 Item Coding and Marginal Fidelity

All source responses use 1 for completely disagree and 5 for completely agree. A421–A424 are traditional statements and are reverse-coded for analysis; A425 is an egalitarian statement and is not reversed. Thus, all reported analytical scores range from 1 to 5, with larger values indicating greater gender egalitarianism.

Table 1: Wave-specific means and total variation under the neutral prompt

Wave	Item	CGSS mean	Qwen mean	Mean error	Total variation
2012	A421	2.519	2.332	$-.187$	.368
	A422	3.007	3.408	.401	.380
	A423	2.884	1.920	$-.964$	.494
	A424	3.867	2.396	-1.471	.707
	A425	3.808	2.202	-1.606	.760
2018	A421	2.974	2.756	$-.218$	.367
	A422	3.263	3.726	.463	.459
	A423	3.100	2.120	$-.980$	.511
	A424	3.985	2.640	-1.345	.617
	A425	3.865	2.472	-1.393	.618
2021	A421	2.908	2.792	$-.116$	.413
	A422	3.251	3.768	.517	.501
	A423	3.116	2.010	-1.106	.524
	A424	4.113	2.500	-1.613	.679
	A425	4.047	2.292	-1.755	.736

Notes: CGSS means use full-sample survey weights. Qwen means pool five draws for each of the 100 sampled profiles per wave. Total variation ranges from 0 for identical category shares to 1 for disjoint distributions.

Table 2: Full-sample prompt ablation

Condition	Profiles	Mean absolute item error	Mean total variation
Neutral verbal-label prompt	300	.942	.542
Original anti-desirability prompt	300	1.170	.664

Notes: Both conditions use the same profiles. The mean same-profile absolute response shift between conditions is .581 points. Because wording and response format change together, this is an ablation rather than a factorial causal contrast.

3 Repeated-Draw and Independent-Item Follow-up

The primary revised analysis uses five joint-prompt draws for every profile. For each profile–item pair, the five observed categories define an empirical predictive distribution. Aggregate moments average those profile-level distributions. The original first draw is retained as a sensitivity analysis.

Total predictive variance is decomposed as

$$\mathrm{Var}(Y) = \mathrm{Var}_i\{\mathrm{E}(Y \mid i)\} + \mathrm{E}_i\{\mathrm{Var}(Y \mid i)\}\,, \tag{1}$$

where the first term is between-profile variance and the second is stochastic variance within a profile. Five draws provide an empirical approximation to both terms.

The independent-item condition makes five separate API calls per profile. Every call repeats the complete persona but contains only one item, starts a new conversation, and contains no previous answer. The direction of the primary contrast was fixed before the full run:

$$\Delta_r = \overline{|r|}_{\text{joint}} - \overline{|r|}_{\text{independent}}.$$

Profile-cluster bootstrap samples retain all items and all repeats for a sampled profile. Profiles are resampled within wave; item and repeat rows are never resampled separately.

4 Heterogeneity and Item Structure

Table 3: Qwen-to-human variance ratios under the neutral prompt

Wave	A421	A422	A423	A424	A425
2012	.412	.577	.256	.660	.305
2018	.592	.367	.389	1.024	.589
2021	.545	.303	.249	.723	.385

Table 4: Recovery of the item-correlation structure

Wave	Correlation-matrix RMSE	Qwen A425 mean $\lvert r\rvert$	CGSS A425 mean $\lvert r\rvert$
2012	.282	.460	.051
2018	.308	.507	.067
2021	.288	.437	.061

Notes: Qwen entries average diagnostics calculated separately for the five repeated draws. A425 mean $|r|$ is the mean absolute Pearson correlation between A425 and A421–A424. Reversing A421–A424 changes correlation signs but not these absolute magnitudes.

Table 5: Repeat stability under the joint neutral prompt

Item	Share identical in all five draws	Mean within-profile SD
A421	.943	.055
A422	.797	.174
A423	.687	.197
A424	.577	.397
A425	.723	.168

5 Monte Carlo Study

Each scenario uses the weighted 2021 CGSS category probabilities. The row bootstrap samples complete respondent rows using survey weights. Independent marginals sample each item separately. The coherent-factor condition uses a common latent normal factor with correlation parameter .65 and empirical category thresholds. Each row in Table 7 summarizes 1,000 samples of $n = 300$.

Table 6: Joint versus independent item presentation

Wave	CGSS mean $\lvert r \rvert$	Joint mean $\lvert r \rvert$	Independent mean $\lvert r \rvert$	Estimable pairs	Joint − independent
2012	.051	.460	.433	2/4	.026
2018	.067	.507	.374	3/4	.133
2021	.061	.437	.308	4/4	.130

Notes: The pooled wave-averaged joint-minus-independent contrast is .096; its 2,000-replication profile-cluster bootstrap interval is $[-.043, .156]$. A422 and A424 are constant or nearly constant in several independent-item cells, so lower correlation is not interpreted as recovered structure.

Table 7: Monte Carlo means and Monte Carlo standard errors

	Abs. mean error		Total variation		Variance ratio		Correlation RMSE		A425 mean $\lvert r \rvert$	
Scenario	Mean	MCSE	Mean	MCSE	Mean	MCSE	Mean	MCSE	Mean	MCSE
Row bootstrap	.0547	.00067	.0426	.00027	.997	.00137	.0505	.00040	.0905	.00118
Independent marginals	.0546	.00061	.0424	.00023	1.000	.00110	.274	.00052	.0457	.00054
Coherent factor	.0559	.00088	.0435	.00028	.999	.00137	.371	.00058	.525	.00083

Notes: MCSE is the replication standard deviation divided by $\sqrt{1000}$. The simulation seed is 20260702.

6 Subgroup-Gradient Recovery

Table 8: Human and Qwen demographic coefficients under the neutral prompt

Wave	Outcome	Predictor	CGSS coefficient	Qwen coefficient	Error
2012	Public index	Female	.089	.359	.270
		Education (years)	.046	.066	.020
		Age (decade)	−.003	−.010	−.007
		Urban residence	.097	.187	.090
	Equal housework	Female	.225	.158	−.067
		Education (years)	.021	.042	.021
		Age (decade)	.013	−.026	−.039
		Urban residence	.000	.003	.003
2018	Public index	Female	.180	.453	.273
		Education (years)	.050	.075	.026
		Age (decade)	−.066	−.003	.064
		Urban residence	.185	.220	.035
	Equal housework	Female	.349	.367	.018
		Education (years)	.024	.084	.061
		Age (decade)	.010	.003	−.008
		Urban residence	−.008	.219	.227
2021	Public index	Female	.174	.389	.215
		Education (years)	.070	.062	−.008
		Age (decade)	−.097	−.036	.062
		Urban residence	.201	.164	−.037
	Equal housework	Female	.328	.130	−.198
		Education (years)	.013	.078	.065
		Age (decade)	−.014	.005	.019
		Urban residence	.053	.128	.075

Notes: Each outcome is regressed simultaneously on all four predictors. Human models use survey weights and full-wave samples; Qwen models use 100 profiles per wave. Across the 24 comparisons, coefficient RMSE is .117 and sign agreement is 83.3 percent.

7 Predictive Benchmarks and Computation

Table 9: Same-profile predictive benchmarks

Model	Ordinal MAE	Abs. mean error	Total variation	Variance ratio
Weighted marginal prior	.975	.127	.097	.992
Multinomial logit	.935	.112	.081	.982
Histogram gradient boosting	.923	.095	.071	.987
Qwen3-8B, neutral prompt	1.379	.917	.538	.466

Notes: All models are evaluated on the same 300 profiles. Machine-learning predictions are out of fold; Qwen metrics use each profile's five-draw empirical distribution. The paired Qwen-minus-HGB absolute-error difference is .456, with a 2,000-replication profile-cluster bootstrap interval of [.407, .507].

Table 10: Temporal transfer: train on 2012 and 2018, evaluate 2021

Model	Ordinal MAE	Total variation	Variance ratio
Weighted historical prior	.997	.113	.960
Multinomial logit	.961	.139	.928
Histogram gradient boosting	.945	.107	.959
Qwen3-8B on matched 2021 profiles	1.442	.561	.459

Notes: Every row evaluates the exact 100-profile 2021 cohort, allowing item-specific missing human answers. The first three rows use only 2012–2018 outcomes for training; Qwen is zero-shot. The comparison concerns reconstruction on a common target cohort, not equal training budgets.

Table 11: Model-call quality and software environment

Component	Recorded value
LLM	Qwen3-8B, non-thinking mode, local Ollama 0.31.1
Decoding	Temperature .7; top-p .9; fixed recorded seed
Hardware	Apple M4 computer; 24 GB unified memory
Calls	3,331 logged attempts; 3,330 successful; success rate 99.97%
Latency	Median 4.92 seconds; 90th percentile 9.66 seconds
Reference models	scikit-learn 1.7.2; five-fold out-of-fold evaluation
R environment	R 4.5.2; package versions recorded in `environment/R-session-info.txt`
Bootstrap	2,000 profile-cluster replications within wave
Monte Carlo	1,000 replications; $n = 300$; seed 20260702
Human sampling calibration	1,000 samples per wave; $n = 100$; seed 20260706

8 Sampling Calibration and Joint Reconstruction

9 Interpretation Boundaries

The revised neutral condition contains five independently seeded draws per profile. The share identical across all five draws ranges from .577 for A424 to .943 for A421. Repetition therefore reveals nonzero stochastic variation, but the resulting total variance remains well below the human benchmark in most wave-item cells.

The `original_verbal` and `paraphrased` conditions contain only ten profiles each. They are parsing diagnostics and are not used as population evidence. Only the 300-profile `original` and `neutral_verbal` conditions support the prompt-ablation results.

Finally, the supervised references and joint donor observe survey outcomes during training, whereas Qwen3-8B does not. Their role is to test whether sparse profile information is a sufficient explanation for

Table 12: A425 association under alternative human correlation estimators

Estimator	2012	2018	2021
Unweighted Pearson	.051	.067	.061
Survey-weighted Pearson	.049	.078	.078
Unweighted polychoric	.069	.091	.084

Notes: Entries are the mean absolute correlations between A425 and A421–A424. The weak human association survives weighting and an ordinal latent-response estimator.

Table 13: Joint outcome-informed benchmark versus Qwen3-8B

Generator	OMAE	TV	Variance ratio	Correlation RMSE	A425 mean $\|r\|$
Stratified joint donor	.987	.070	.973	.085	.105
Qwen3-8B	1.379	.538	.466	.293	.468

Notes: Donors are matched on wave, sex, education group, and urban residence; the matched respondent is excluded. Five complete response vectors are sampled per profile using survey weights. OMAE, TV, and variance use the same matched profiles as the primary benchmark. Relational metrics compare each generated 100-profile matrix with the weighted full-sample correlation matrix.

Table 14: Wave-specific human sampling envelopes and Qwen3-8B estimates

Wave	Metric	Human 2.5th	Human 97.5th	Qwen
2012	Absolute mean error	.036	.160	.926
	Total variation	.048	.102	.542
	Variance ratio	.855	1.137	.442
	Correlation RMSE	.051	.137	.282
	A425 mean $\|r\|$	.037	.198	.460
2018	Absolute mean error	.036	.163	.880
	Total variation	.048	.100	.514
	Variance ratio	.856	1.141	.592
	Correlation RMSE	.052	.141	.298
	A425 mean $\|r\|$	.041	.234	.507
2021	Absolute mean error	.036	.168	1.021
	Total variation	.046	.104	.571
	Variance ratio	.846	1.138	.441
	Correlation RMSE	.047	.137	.279
	A425 mean $\|r\|$	.034	.219	.437

Notes: Human limits come from 1,000 repetitions of the same stratified, probability-proportional-to-weight profile design with $n = 100$ per wave. Qwen lies outside the central 95 percent human reference envelope in all 15 cells.

the recorded distortion after outcome calibration. They do not isolate architecture from calibration, establish that a supervised learner is a substitute for human respondents, or model individual psychology.